Expandable gears transmission

PRIOR ART

[0001] Transmissions that transfer rotational power of a prime mover to a driven mover with speed and torque varied are widely utilized in transportation, manufacture, agriculture and other industries. Common designs known in the art include continuously variable transmissions (CVTs), gearboxes and bicycle derailleur-based drivetrains.

[0002] Several commonly considered factors determine the suitability of a transmission design for a specific practical application. These factors include gearing range, maximum torque, size, weight, efficiency, durability, cost, and maintenance requirements.

[0003] For example, a drivetrain of a pedal powered bicycle needs to withstand relatively high torque loads as, compared to a transmission in a motor vehicle, it rotates very slowly. CVT and gearbox transmission designs that support such torque loads are relatively heavy and expensive to manufacture. Consequently, parallelogram derailleur-based drivetrains dominate the multi-speed bicycle market.

[0004] Bicycle derailleur drivetrains typically feature a 6 to 12 speed cassette mounted to the rear wheel hub and 1 to 3 chainrings mounted to the bottom bracket. Chain is utilized to transfer power between front and rear mechanisms. In most gears, the engaged cassette sprocket and a chainring are not perfectly aligned in a so-called “ideal chain line”. This misalignment forces the chain to bend laterally, resulting in increased friction and wear.

[0005] Several approaches have been proposed to address this issue by modifying the sprocket size in place, rather than shifting the chain to sprockets outside the ideal chain line. The following patents describe different adjustable sprocket mechanisms, outlining the current state of the art and highlighting problems and challenges. U.S. Pat. No. 4,493,678, by Royce H. Husted, filed on Jun. 11, 1982, describes a design entitled Expandable Sprocket. U.S. Pat. No. 4,740,190, by Joseph A. Pike, filed on Jan. 29, 1987, describes a design entitled Variable Ratio Bicycle Chain Drive. U.S. Pat. No. 4,850,939, by Keith C. Chilcote and Robert J. Mahlowski, filed on Oct. 21, 1987, describes a design entitled Variable Diameter Sprocket Assembly. U.S. Pat. No. 4,952,196, by Keith C. Chilcote and Robert J. Mahlowski, filed on Oct. 21, 1987, describes a design also entitled Variable Diameter Sprocket Assembly. U.S. Pat. No. 5,984,814, by Bruce Davenport, filed on Sep. 4, 1997, describes a design entitled Variable Ratio Chain Drive Transmission. These solutions introduce additional drawbacks that the present invention avoids.

[0006] Next, differences between sprocket sizes in lowest and highest gear results in so-called “chain slack”. A derailleur cage with a dual pulley arrangement is required to maintain chain tension. To manage substantial chain slack in the harder gears, a profound bending of the chain is required. This increases friction in the system.

[0007] Single-speed bicycles and bicycles equipped with multi-speed gearboxes or gear hubs often utilize a rubber or carbon belt for power transfer to the rear wheel. Belts are quieter than chains and do not require lubrication, thus reducing maintenance needs. However, common drivetrain belts do not offer enough lateral flexibility to be compatible with derailleur drivetrains.

[0008] It would be a significant advancement in the field to provide a transmission that enhances the efficiency and longevity of conventional derailleur drivetrains by maintaining an ideal chain line, while also being compatible with belt drives and not compromising other

performance factors. Such a transmission would be suitable for a variety of applications, including bicycle drivetrains.

BACKGROUND OF THE INVENTION

[0009] This invention relates to belt or chain power transmission systems, specifically to a gear set with sprockets composed from multiple components.

SUMMARY OF THE INVENTION

[0010] A transmission system based on magnetic pins in place of traditional sprocket teeth. An electromagnetic shifting mechanism moves selected pins between engaged and idle positions thus forming sprockets of varying sizes that drive a belt or chain.

[0011] The invention is suitable for applications that value high efficiency and low weight. It is intended to be utilized in, but not limited to: road bicycles, mountain bicycles, motor-assisted bicycles, e-bikes, tricycles, recumbent bicycles, and motor vehicles.

[0012] Primary application of the invention is to replace the common bicycle drivetrain, namely the front and rear derailleurs, chain, cassette and chainrings.

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. 1 shows a right side perspective view of one example of a bicycle equipped with two transmission units described in the present disclosure.

[0014] FIG. 2 shows a right side perspective view of the rear transmission unit of FIG. 1.

[0015] FIG. 3 shows a top view of the rear transmission unit without the belt.

[0016] FIG. 4 shows a right side perspective view of the rear transmission without the drive plate, while the smallest gear engaged.

[0017] FIG. 5 shows the transmission unit of FIG. 4 without the axle, during a multi-shift to larger gear.

[0018] FIG. 6 shows a right side view of the transmission unit of FIG. 5.

[0019] FIG. 7 shows the transmission unit of FIG. 6 after the multi-shift was completed.

[0020] FIG. 8 shows the transmission unit of FIG. 6 after a shift to the largest gear.

[0021] FIG. 9 is a detailed view of a transmission cross-section showing a pin in the engaged position.

[0022] FIG. 10 is a detailed view of a transmission cross-section showing a pin in the idle position.

[0023] FIG. 11 shows a perspective view of an alternate pin profile and a belt.

[0024] FIG. 12 shows a side view of FIG. 11.

DETAILED DESCRIPTION OF THE INVENTION

[0025] The present invention improves some of the characteristics of other transmission systems. In a bicycle application, it provides a higher efficiency than a derailleur-based drivetrain as it avoids above-noted sources of friction. It is belt drive compatible which reduces maintenance requirements and noise. Other important characteristics, such as gearing range, maximum torque, size, weight and durability remain similar. While complexity is higher than that of a mechanical derailleur drivetrain, it is comparable to state-of-the-art wireless electronic drivetrains.

[0026] Another advantage is that the width of the transmission unit is constant, regardless of how many gears it features. It is comparable to 3 traditional bicycle sprockets. Thus, a twelve gear transmission unit will be about four times narrower than a typical twelve speed bicycle cassette, and can replace the conventional front derailleur of a bicycle, which commonly doesn't feature more than three chainrings.

[0027] Alternatively, the transmission unit can replace the conventional rear derailleur and cassette of a bicycle. A smaller overall width allows a wider spoke placement, and thus a stronger rear wheel. This setup is very similar to a conventional single chainring (1x) drivetrain and is compatible with many existing bicycle frames and components.

[0028] Finally, two transmission units can be utilized on a single bicycle. This arrangement has several advantages over a single unit configuration. For example, a front unit with six gears mated to a seven-gear rear unit will feature twelve effective gears when shifted in alternating pattern. The units will be smaller than of a single twelve-gear unit with similar range, the total number of teeth (magnetic pins) will be reduced, and a shorter belt with a simple tensioner will suffice.

[0029] The decision on whether to use a single rear-mounted unit, a single front-mounted unit, or a dual unit depends on a particular bicycle type. Those having ordinary skill in the art should appreciate the benefit of unsprung weight reduction resulting from the use of a single front-mounted unit on a dual-suspension mountain bike, or the simplicity of adapting a common bicycle with a rear-mounted unit, or the aerodynamic and gear range advantage of a dual unit configuration on a road bicycle.

[0030] Turning now to the drawings, FIG. 1 depicts one example of a bicycle 100 with a frame 101, a front wheel 102 coupled to a fork 103 of the frame, and a rear wheel 104. A front gear assembly 110 is fixedly connected to cranks 114. A belt 116 transmits power from the front gear assembly 110 to the rear gear assembly 112, which is fixedly connected to the rear hub 118. A spring loaded belt tensioner 130 is connected to the frame 101. Shifting mechanisms 140 and 142 form a solid connection with the frame 101 and thus do not rotate. The front gear assembly 110 and the shifting mechanism 140 form a front transmission unit 150. The rear gear assembly 112 and the shifting mechanism 142 form a rear transmission unit 152.

[0031] A single electronic shift controller 105 features upshift and downshift buttons, and is of a common design known in the art.

[0032] The shifting mechanisms 140 and 142 and the shift controller 105 are connected either by wired or wireless data connection for two-way communication.

[0033] FIGs. 2-8 show substantial portions of the rear-mounted transmission unit 152 of FIG. 1. Main plate 200 and drive plate 210 are driven in a clock-wise direction by the belt 116. The rear hub 118 is a conventional bicycle hub variant commonly known as a freehub. It is connected to frame 101 with axle 119.

[0034] Drive plate 210 is not shown in FIG. 4 to reveal the fixed, and currently engaged smallest sprocket 230. This seven speed transmission variant features ninety-nine identical pins 240, currently fully retracted within the main plate 200 in an idle position. The pins 240 can form six virtual sprockets when engaged as shown on FIG. 8.

[0035] Shifting mechanism 120 features two electromagnets on each side of each virtual sprocket (total of twelve electromagnets for this transmission variant with seven gears). Each of the electromagnets supports three states – an off state, south pole towards the drivetrain, and

north pole towards the drivetrain. The second and third states are accomplished by reversing the flow of current through the electromagnet. Alternatively, the shifting mechanism can consist of permanent magnets rotated by small electric motors. Use of a single-sided shifting mechanism with stronger electromagnets or magnets is also a possibility.

[0036] Pins 240 are of a cylindrical shape, and constructed from a permanently magnetized metal with magnetic poles on either end. Main plate 200 and drive plate 210 are constructed from a non-ferrous material, such as a self-lubricating plastic polymer. Every pin 240 can be fully inserted in its housing 250 on the main plate. This is referred to as an idle pin position. FIG. 11 shows a cross-section of a single pin 240 in the idle position and on FIG. 4 all pins are in the idle position. Conversely, housings on the drive plate 210 only accept a tip of each pin. This is referred to as a drive pin position. FIG. 10 shows a cross-section of a drive plate 210, main plate 200 and a single pin 240 in the engaged position and on all pins on FIG. 8 are in the engaged position. Attached near the bottom of every pin housing are small plates 202 and 212 from a material such as a soft metal, that can become a temporary magnet. Thus, when the shifting mechanism 120 is the off position, all pins are secured in either idle or drive position with magnetic force between the pin 240 and small plates 202 or 212 respectively.

[0037] The shifting mechanism 120 requires a power source. This can be in the form of a dedicated, removable, rechargeable direct current battery (not pictured). As the shifting mechanism is fixedly connected to the bicycle frame, a wired connection to a shared power-source, such as an e-bike powertrain battery, can be utilized.

[0038] A gear change can occur when the fixedly connected rear hub 118, main plate 200, drive plate 210 and the smallest sprocket 230 rotate on the axle 119. FIG. 4 shows the transmission state just before a multi-shift from the smallest sprocket 230 (seventh gear) to the third virtual sprocket (fifth gear, composed from pins) is initiated. The method of initiating the shift can be implemented in various ways; in this embodiment the bicycle user presses the upshift button on a common electronic shifter controller. Alternatively, an automated system can monitor the state of the transmission and initiate a shift when required. The shift instruction is propagated to the shifting mechanism 120 which engages electromagnets for sprocket two three – 270, 271, 277 and 278.

[0039] FIG. 5 and FIG. 6 show the transmission unit of FIG. 4 after it rotated 180 degrees clockwise since the shift was initiated. As the drivetrain rotated, pins 320 to 325 of the second sprocket passed within a small distance of their respective electromagnets 270 and 277, and pins 330 to 335 of the third sprocket passed within a small distance of electromagnets 271 and 278. Interaction of the magnetic material of the pins with their respective electromagnets moved the pins to their engaged positions. Belt 116 was forced to travel on top of the engaged pins 320 and 330. Eventually, the belt's grooves partly engaged up with pin 331, and then fully engaged with pin 332.

[0040] As the gear assembly will continue to rotate, pins 333, 334 a 335 will engage with the belt 116. Remaining idle pins for the third sprocket, starting with the pin 336, will be moved to engaged position by magnets 271 and 278, one by one.

[0041] FIG. 7 shows the transmission unit of FIG. 4 after a full 360 degrees rotation. The shift to a larger sprocket is complete, and the belt 116 is fully engaged with pins 330 to 339.

[0042] A shift to a smaller sprocket is analogical with the sole difference that sprocket pins are moved from their engaged positions into their opposite, idle positions by the respective electromagnets.

[0043] FIG. 8 shows the transmission unit in the largest gear with all pins 240 in their engaged positions.

[0044] To summarize, some common characteristics of a single transmission unit are as follows:

[0045] It is possible to initiate an up-shift or down-shift by any number of gears.

[0046] It is not necessary for a shift to complete before a different shifting instruction can be accommodated; for example, in the middle of a shift from a smallest to largest gear, the operator can change their mind and adjust the shift to only the middle gear.

[0047] The transmission can operate in both directions. Thus, in the presented bicycle application, the operator can optionally be allowed to shift to the desired gear by pedaling backwards, while the bicycle is stationary.

[0048] The shift controller of the transmission can be instructed to prioritize belt engagement over shifting speed or vice versa and thus tune it to suit a particular application.

[0049] Standard bicycle chains are not compatible with the invention. A toothed belt, as shown in FIGs. 1-8, is ideal for most applications. Alternatively, an inverted tooth chain, which is known in the art, can be utilized.

[0050] It is not necessary for the smallest sprocket 230 to be a standard fixed sprocket. It could be composed from pins similarly to the other virtual sprockets, only such that are permanently engaged in the drive position. Alternatively, the smallest sprocket could be mounted on the shaft in such a manner that it can spin freely, rather than be fixedly connected. A bicycle equipped with such a fully disengaged gear would allow it to be operated in a backwards motion with stationary cranks and pedals. This is desirable for some stunt bicycles.

[0051] Referring now back to FIG. 1 to further elaborate on the operation of a dual transmission setup. In this embodiment, the front transmission unit 150 is a six gear variant with tooth counts of 30, 25, 22, 19, 16, 14. The rear transmission unit 152 features seven gears with tooth count of 9, 11, 13, 15, 17, 20, 23. Smallest sprockets are traditional, other sprockets are composed from pins as detailed above. Together they form a system with twelve effective gears and a 548% range.

[0052] A single shift controller can operate both units synchronously. The shifting pattern alternates the units; starting from the lightest gear 14:23 (front:rear), the next gear is a rear unit shift to 14:20, then a front unit shift to 16:20, followed by rear unit shift to 16:17, and so on, the full pattern being 19:17, 19:15, 22:15, 22:13, 25:13, 30:11, 30:9.

[0053] This pattern minimizes “chain slack” (the difference between maximum and minimum engaged tooth count) to only 8 teeth, thus enabling the use of a simple and efficient single-pulley belt or chain tensioner 130. A single unit setup with a similar range would require a large gear with approximately 49 teeth resulting in a larger footprint and weight. Such a setup would require a longer belt or chain and a more complex tensioner 130.

[0054] Additionally, both single shifts and multi-shifts of the dual transmission setup will often take less time to complete, as, in most gears, the belt or chain will spin faster than it would in a comparable single unit setup.

[0055] In FIGs. 1-10 all pins 240 are displayed as having a simple cylindrical shape. This circle profile can be further optimized, along with the belt 116 profile, for an improved belt engagement and thus higher supported torque as shown in FIGs. 11 and 12.

[0056] The illustrations of the embodiments described herein are intended to provide a general understanding of the structure of the various embodiments. The illustrations are not intended to serve as a complete description of all of the elements and features of apparatus and systems that utilize the structures or methods described herein. Many other embodiments may be apparent to those of skill in the art upon reviewing the disclosure. Other embodiments may be utilized and derived from the disclosure, such that structural and logical substitutions and changes may be made without departing from the scope of the disclosure. Additionally, the illustrations are merely representational and may not be drawn to scale. Certain proportions within the illustrations may be exaggerated, while other proportions may be minimized. Accordingly, the disclosure and the figures are to be regarded as illustrative rather than restrictive.

[0057] While this specification contains many specifics, these should not be construed as limitations on the scope of the invention or of what may be claimed, but rather as descriptions of features specific to particular embodiments of the invention. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

[0058] Similarly, while operations and/or acts are depicted in the drawings and described herein in a particular order, this depiction should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments.

[0059] One or more embodiments of the disclosure may be referred to herein, individually and/or collectively, by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any particular invention or inventive concept. Moreover, although specific embodiments have been illustrated and described herein, it should be appreciated that any subsequent arrangement designed to achieve the same or similar purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all subsequent adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, are apparent to those of skill in the art upon reviewing the description.

[0060] The Abstract of the Disclosure is provided with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, various features may be grouped together or described in a single embodiment for the purpose of streamlining the disclosure. This disclosure is not to be interpreted as reflecting

an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may be directed to fewer than all of the features of any of the disclosed embodiments. Thus, the following claims are incorporated into the Detailed Description, with each claim standing on its own as defining separately claimed subject matter.

[0061] It is intended that the foregoing detailed description be regarded as illustrative rather than limiting and that it is understood that the following claims including all equivalents are intended to define the scope of the invention. The claims should not be read as limited to the described order or elements unless stated to that effect. Therefore, all embodiments that come within the scope and spirit of the following claims and equivalents thereto are claimed as the invention.

Claims

1. A gear assembly, comprising: main and drive plates coupled to a crank assembly for rotation therewith about a rotation axis, pins that can move between main and drive plates to form additional sprockets, an optional fixed sprocket smallest gear.
2. A transmission unit consisting of a gear assembly of claim 1, and an electromagnetic shifting mechanism that controls the gear assembly of claim 1.
3. A transmission unit of claim 2 that transmits power to or from a continuous toothed belt, with a pin profile optimized for the belt profile.
4. A transmission unit of claim 2 that transmits power to or from a continuous inverted tooth chain with a pin profile optimized for the chain profile.
5. A dual transmission assembly, consisting of two transmission units of claim 2, a tensioner, and connected by a belt or chain.
6. A dual transmission assembly of claim 5, controlled by a single electronic shifter, and shifting in an alternating manner.

Abstract

A transmission including sprockets formed from magnetic pins in place of traditional sprocket teeth, and a common small sprocket. While in operation, a stationary electromagnetic shifting mechanism moves pins between engaged and idle positions thus forming sprockets of varying diameter and tooth count. The largest engaged sprocket drives a toothed belt or an inverted tooth chain. The transmission can be located on the prime mover shaft, the driven shaft, or on both shafts and can operate both in a clockwise and anti-clockwise direction. It features two gears at a minimum. Maximum number of gears is unlimited. Additional gears increase the diameter, complexity, and weight of the unit, while width remains constant. The transmission is suitable for applications that value high efficiency and low weight, primarily as a replacement for the common bicycle drivetrain.

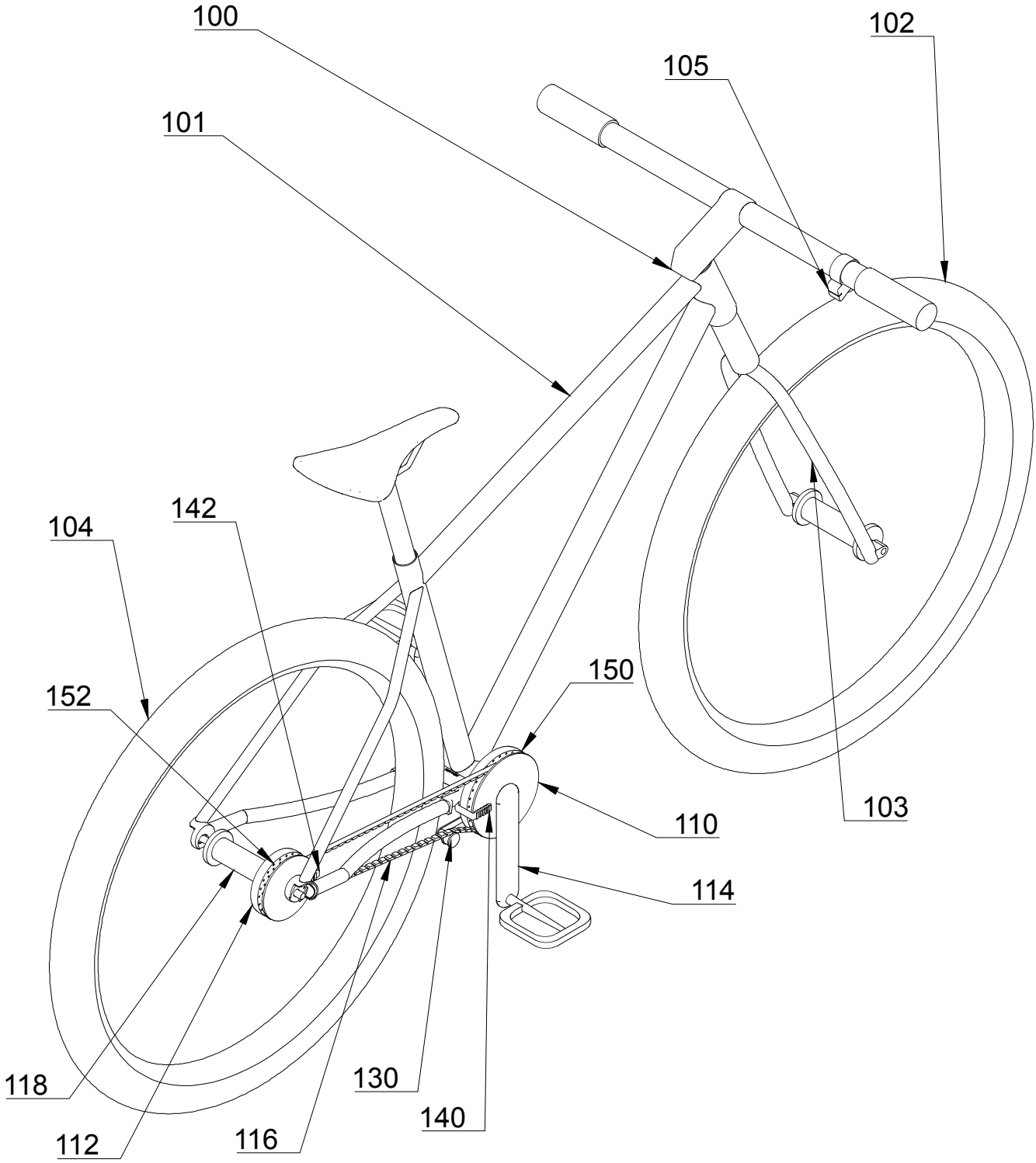


FIG. 1

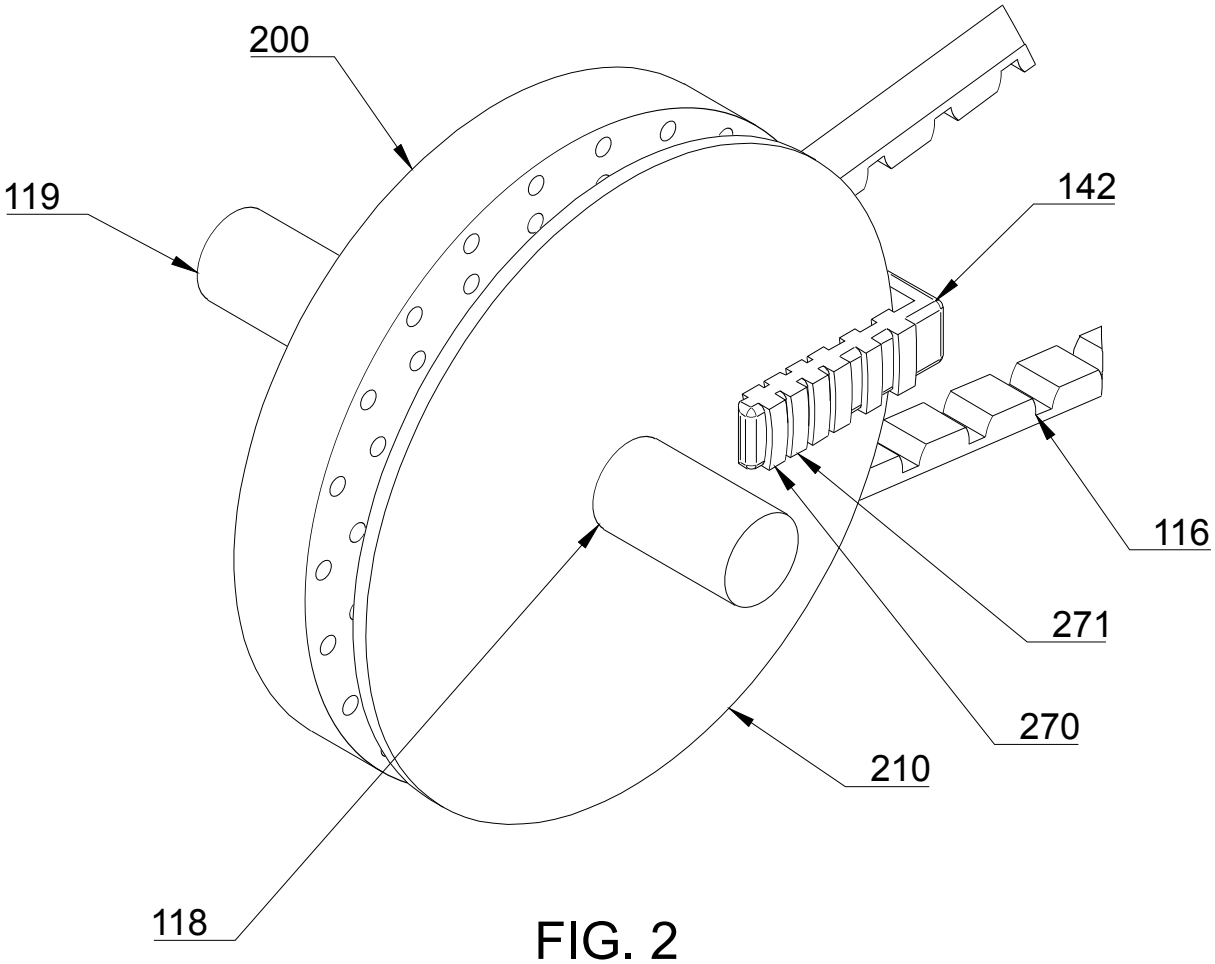

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FIG. 2

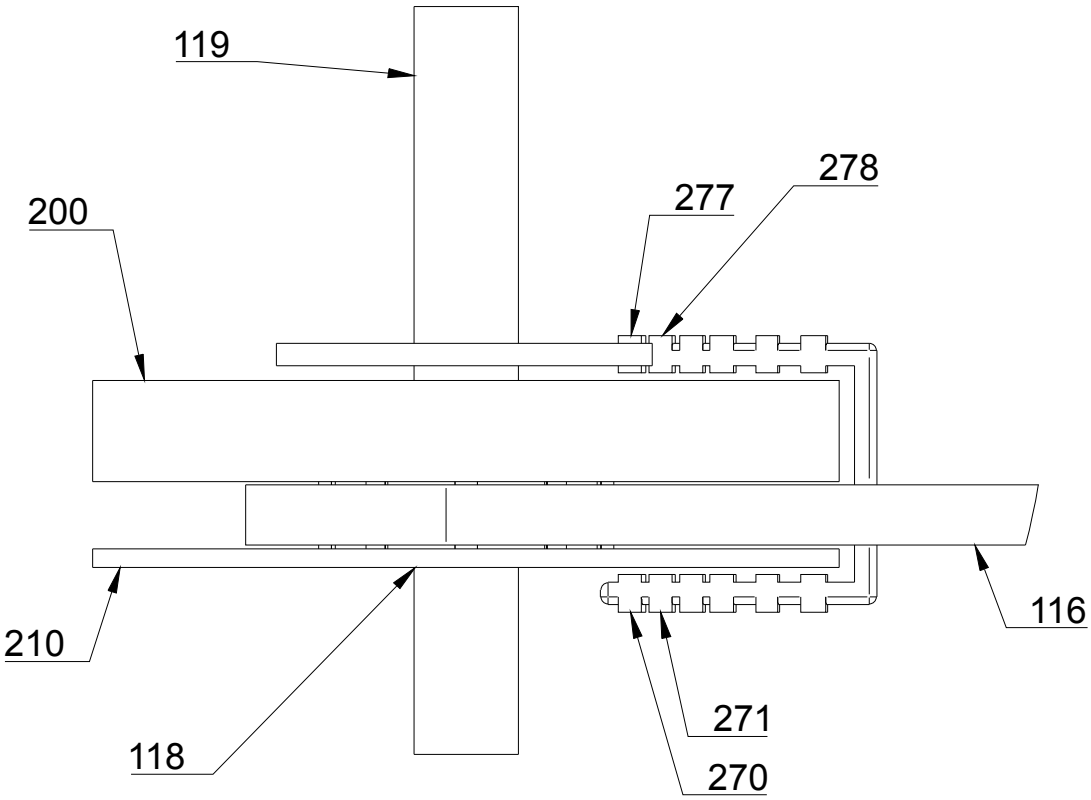

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FIG. 3

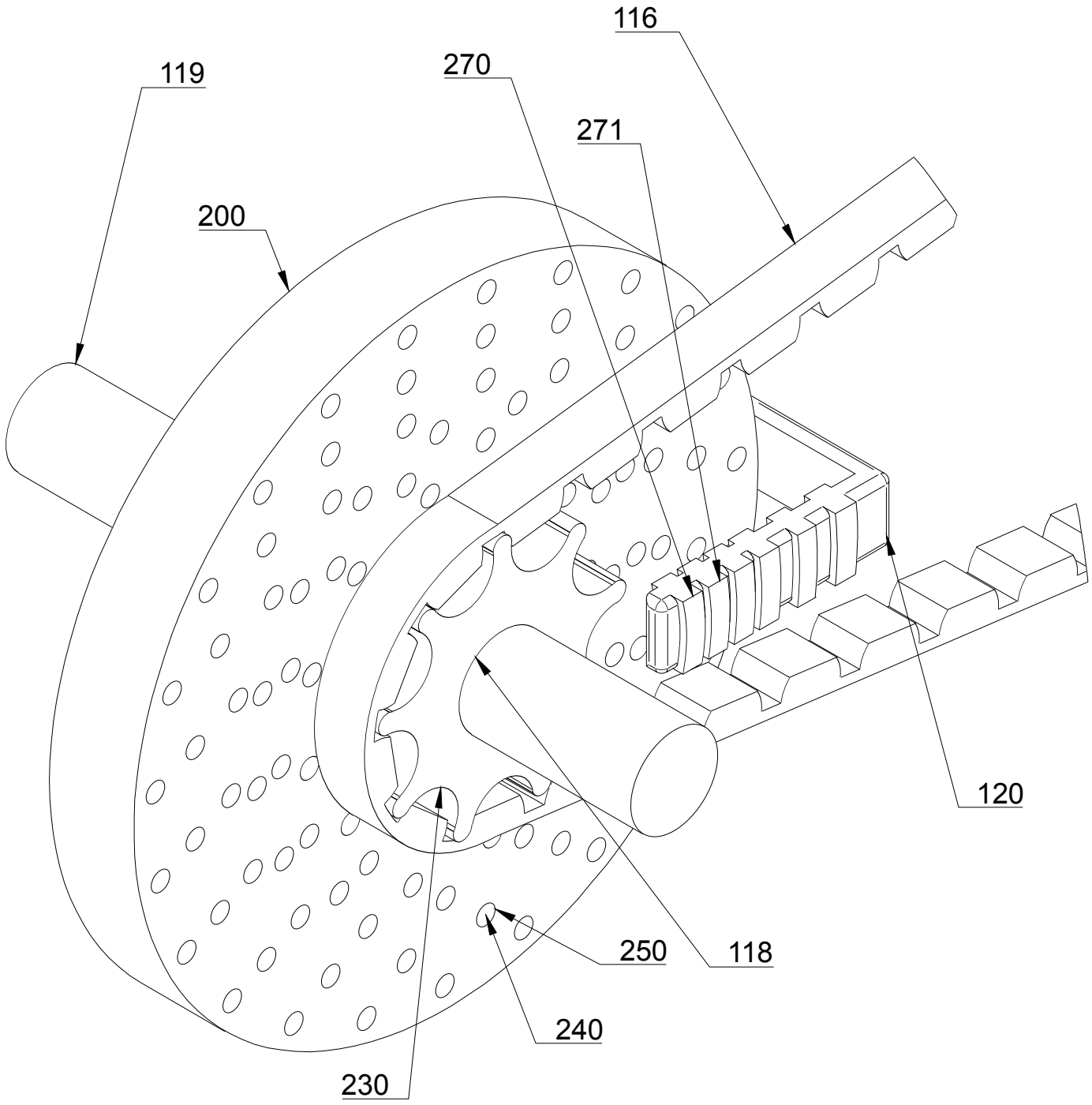


FIG. 4

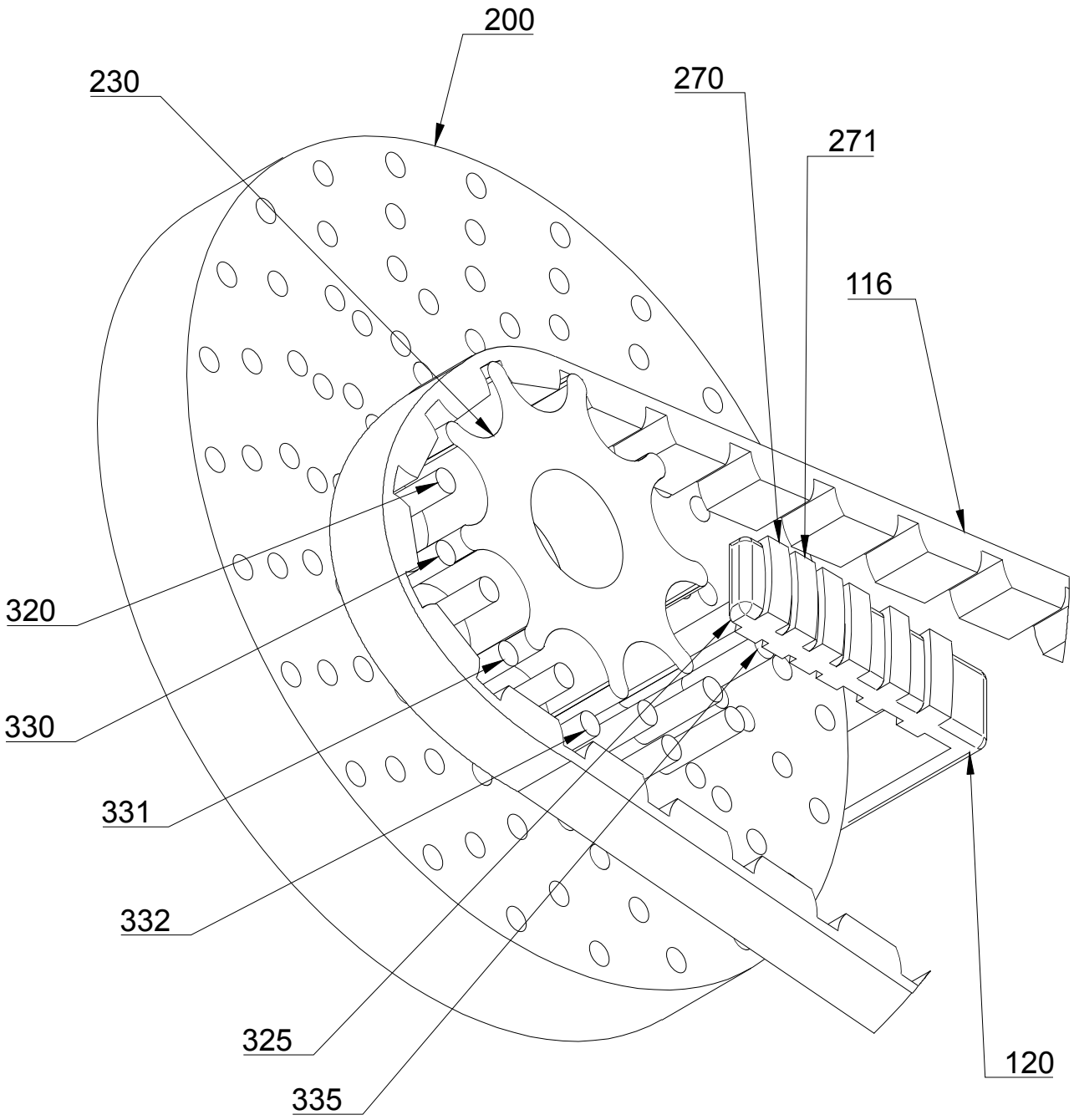


FIG. 5

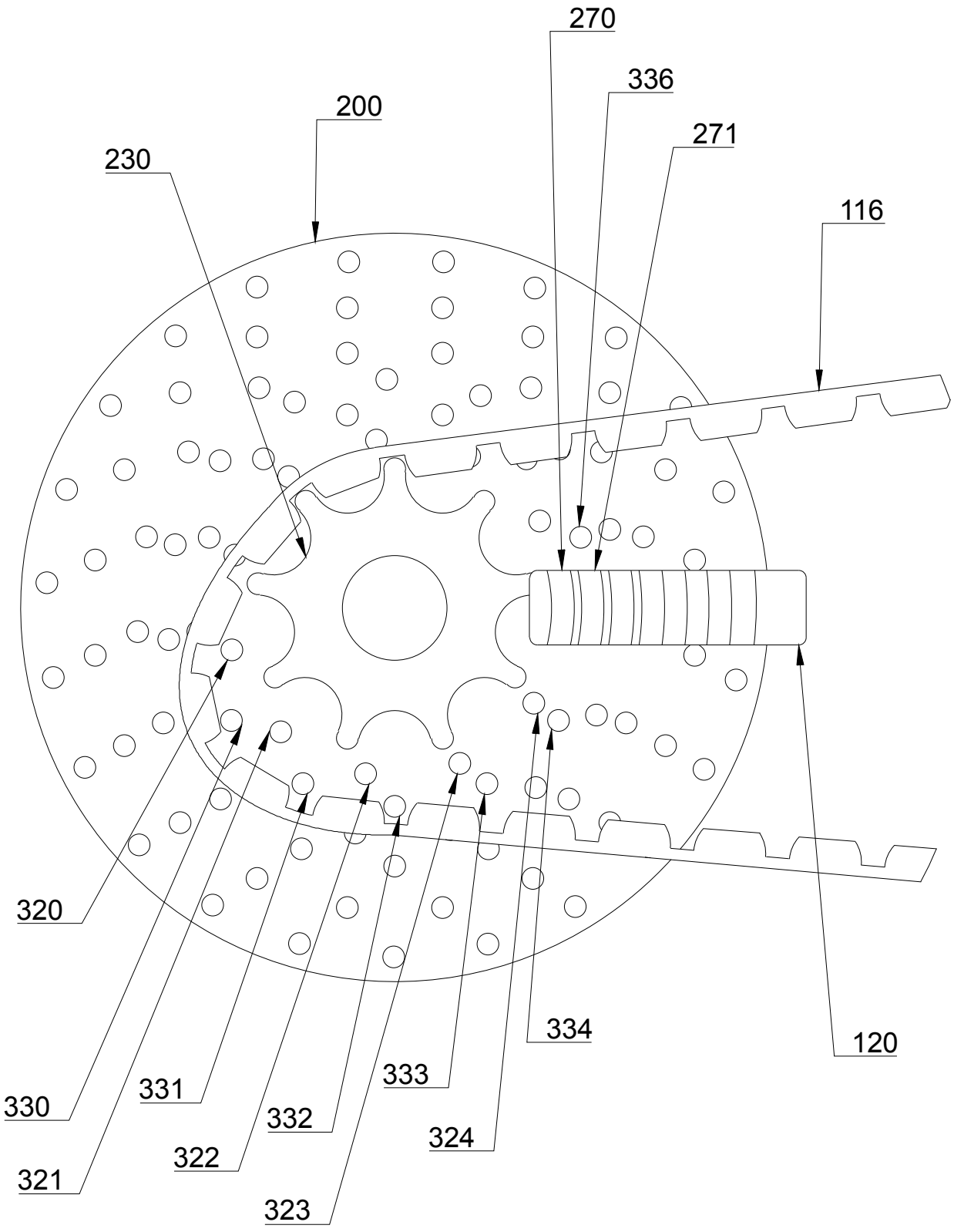

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FIG. 6

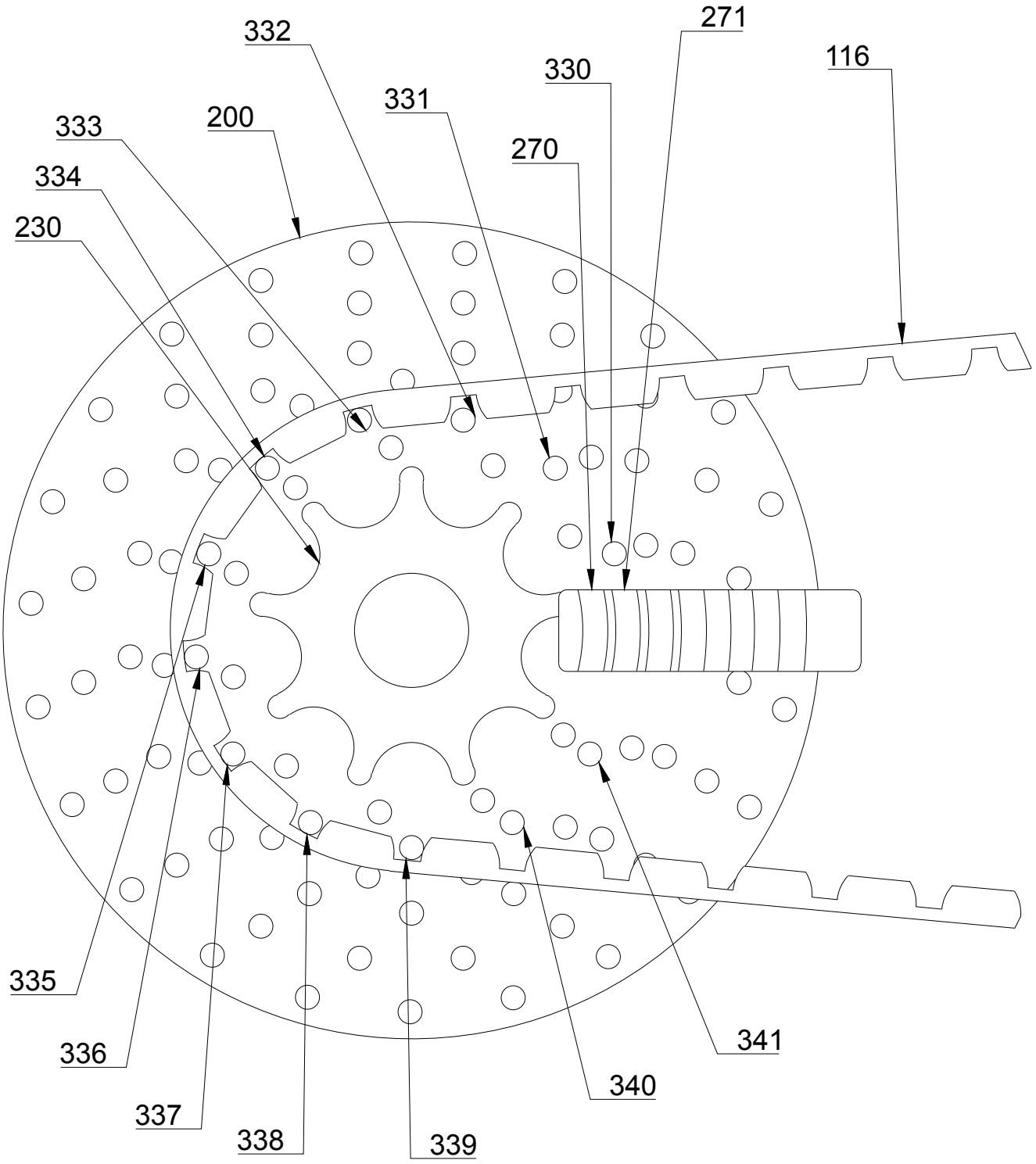

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FIG. 7

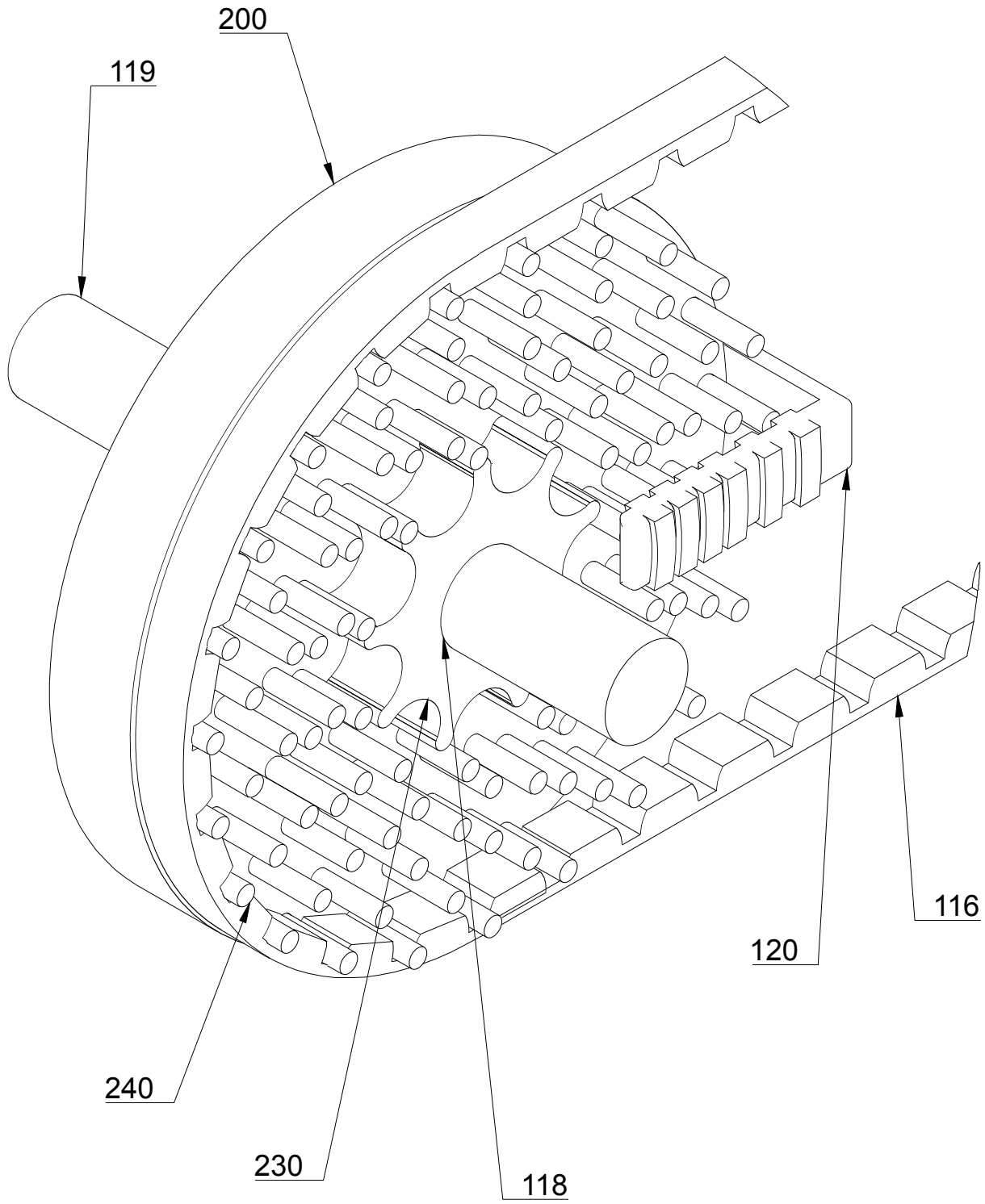


FIG. 8

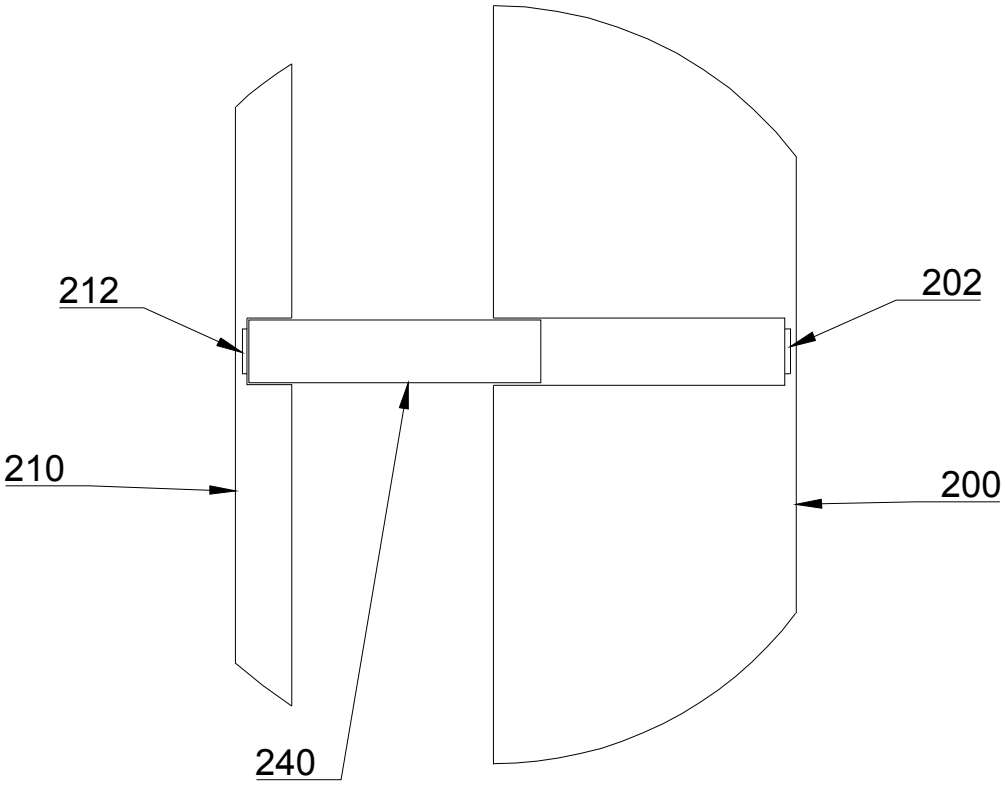


FIG. 9

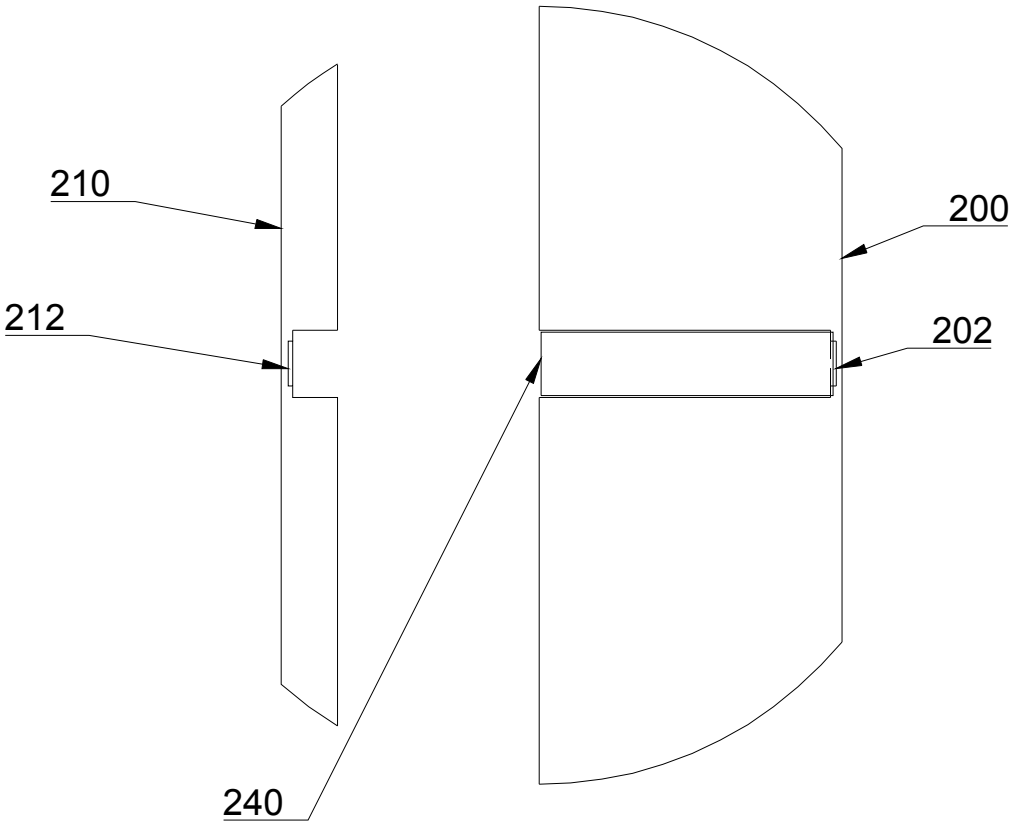


FIG. 10

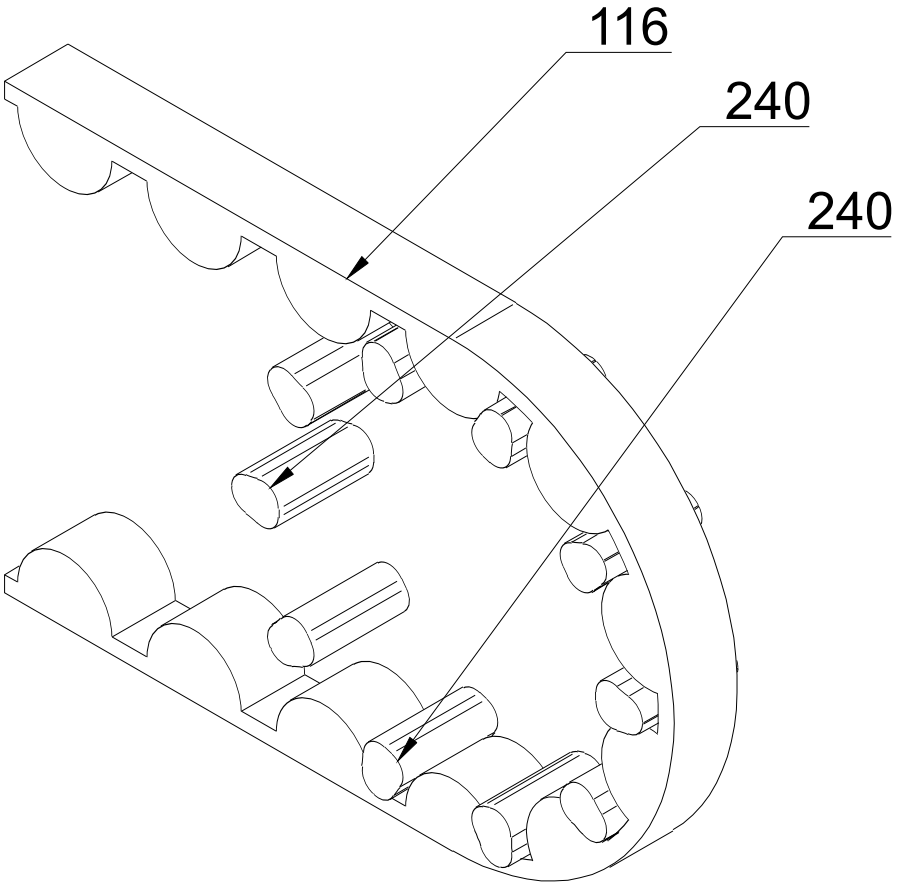


FIG. 11

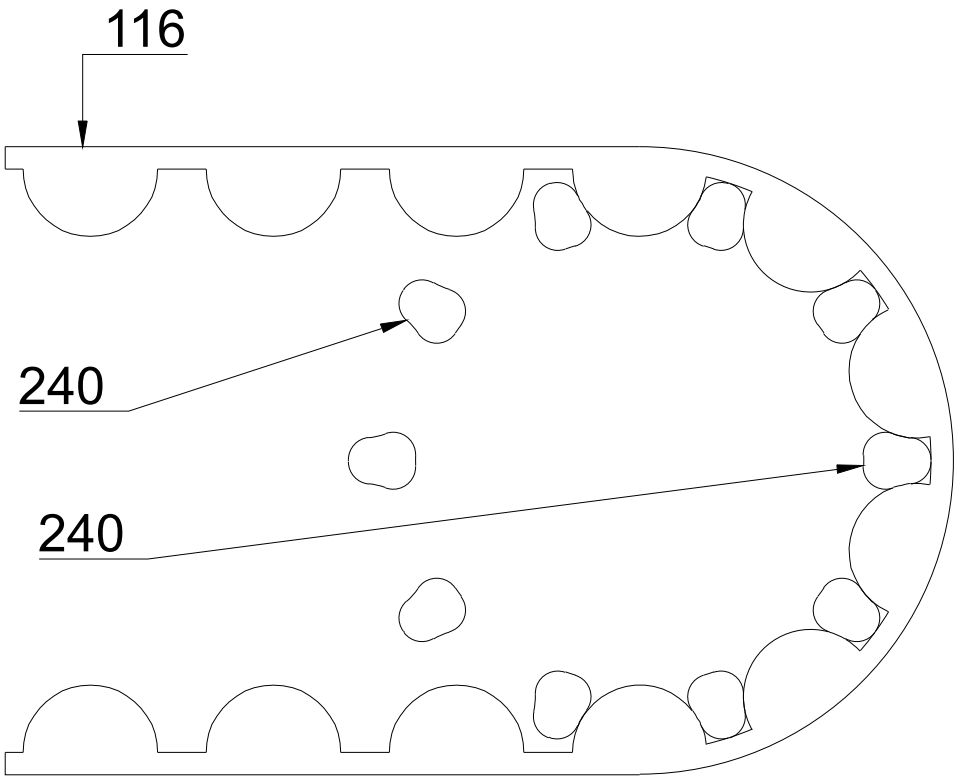


FIG. 12